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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

PH2501-1 – NANOTECHNOLOGY
(Open Elective)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **A nanometer is equivalent to:**
A) 10^{-3} meters
B) 10^{-6} meters
C) 10^{-9} meters
D) 10^{-12} meters
2. **Which of the following is NOT a common classification of nanomaterials?**
A) Zero-dimensional
B) One-dimensional
C) Two-dimensional
D) Four-dimensional
3. **The number of crystal systems in crystallography is:**
A) 4
B) 5
C) 6
D) 7
4. **In Miller indices, the notation () refers to:**
A) A direction
B) A plane
C) A point
D) A crystal system
5. **The energy gap between the valence band and the conduction band in semiconductors is typically:**
A) 0 eV
B) ~2 eV
C) ~5 eV
D) ~10 eV
6. **Compared to their bulk counterparts, nanomaterials often exhibit**
A) Lower strength
B) Higher strength
C) Same strength
D) Variable strength
7. **The ability of a material to store electrical energy in an electric field is termed:**
A) Conductivity
B) Resistivity
C) Permeability
D) Permittivity
8. **As the size of a material decreases to the nanoscale, its surface-to-volume ratio**
A) Decreases
B) Increases
C) Remains the same
D) Becomes negligible
9. **The Scherrer equation is used to determine:**
A) Crystallite size
B) Crystal lattice parameters
C) Interplanar spacing
D) Crystal symmetry
10. **In SEM, images are formed by detecting:**
A) Transmitted electrons
B) Secondary electrons
C) X-rays
D) Auger electrons
11. **Which of the following is a key advantage of TEM over SEM?**
A) Ability to image surface topography
B) Higher depth of field
C) Higher resolution
D) Larger sample size compatibility

12. Which technique is NOT a type of Scanning Probe Microscopy?
 A) Atomic Force Microscopy B) Scanning Tunneling Microscopy
 C) Scanning Electron Microscopy D) Piezoresponse Force Microscopy
13. In AFM, the interaction between the probe and the sample is measured by detecting:
 A) Electron emission B) Photon emission
 C) Cantilever deflection D) X-ray fluorescence
14. XPS is a technique used to analyze:
 A) Bulk material composition B) Surface elemental composition
 C) Magnetic properties D) Optical properties
15. Photoluminescence spectroscopy measures:
 A) Light absorption B) Light emission
 C) Thermal emission D) Magnetic resonance
16. UV-VIS-IR spectrophotometers are used to measure:
 A) Magnetic properties B) Optical absorption and transmission
 C) Electrical conductivity D) Thermal conductivity
17. Which of the following is NOT an allotrope of carbon?
 A) Graphene B) Diamond
 C) Graphite D) Silicon carbide
18. Which crystal structure is characteristic of a diamond?
 A) Hexagonal close-packed B) Body-centered cubic
 C) Face-centered cubic D) Simple cubic
19. What is a primary advantage of using nanostructured thin films in material science?
 A) Increased weight B) Enhanced surface area
 C) Reduced chemical reactivity D) Decreased mechanical strength
20. How are quantum dots primarily used in the field of medicine?
 A) As antimicrobial agents B) For bio-imaging and diagnostics
 C) As dietary supplements D) In surgical sutures

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are nanomaterials? Discuss their unique properties compared to bulk materials.	4	L*2	1	1,2
	b) Classify nanomaterials based on their dimensions and provide examples for each category.	6	L2	1	1,2
	c) Discuss the mechanical properties of nanomaterials and how they differ from their bulk counterparts.	6	L2	3	1,2
2.	a) Explain the difference between conductors, semiconductors, and insulators based on their band structures.	4	L2	2	1,2
	b) How are crystal directions and planes determined in a crystalline structure?	6	L3	2	1,2
	c) Describe the dielectric properties observed in nanostructured materials.	6	L2	3	1,2
3.	a) Differentiate between Physical Vapor Deposition (PVD) and Chemical Vapor Deposition (CVD) techniques.	4	L2	3	1,2
	b) Discuss the sol-gel synthesis method and its applications in nanomaterial fabrication.	6	L2	3	1,2
	c) Discuss the various growth techniques employed in fabricating semiconductor nanostructures.	6	L2	3	1,2
Unit – II					
4.	a) Derive Bragg's law.	5	L2	4	1,2

b)	Describe the working principle and construction of scanning electron microscopy and how it differs from optical microscopy.	8	L2	4	1,2
c)	X-rays with a wavelength of 0.154 nm produce a first-order diffraction peak ($n=1$) at a Bragg angle (θ) of 30° . Determine the interplanar spacing (d) in the crystal.	3	L3	4	1,2
5. a)	Highlight the difference between SEM and TEM.	5	L2	4	1,2
b)	Describe the principle and operation of Atomic Force Microscopy (AFM). What types of information can it provide about a sample's surface?	8	L2	4	1,2
c)	Calculate the wavelength of electrons accelerated through a potential difference of 20 kV in SEM. (Mass of the electron ' m ' = 9.1×10^{-31} kg and charge of electron ' e ' = 1.609×10^{-19} C) and $h = 6.626 \times 10^{-34}$ Js.	3	L3	4	1,2
6. a)	Compare infrared and Raman spectroscopy in terms of their principles and the molecular information they provide.	5	L2	4	1,2
b)	Describe the working principle and construction of UV-VIS spectrophotometers and their application in material characterization.	8	L2	4	1,2
c)	A solution exhibits an absorbance of 0.500 at a wavelength where the molar absorptivity (ϵ) is $200 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$. The path length (l) of the cuvette is 1 cm. Calculate the concentration (c) of the solution.	3	L3	4	1,2
Unit – III					
7. a)	What are the distinct structural and electronic properties of various carbon allotropes, such as diamond, graphite, graphene, and carbon nanotubes?	8	L2	5	1,2
b)	What are the primary methods employed in the synthesis of diamond, and how do they influence nucleation and growth processes?	8	L2	5	1,2
8. a)	Explain the applications of nanotechnology in biology and medicine.	6	L2	5	1,2
b)	What are the advantages of using nanostructured thin films in electronic and optoelectronic devices?	5	L2	5	1,2
c)	Discuss the on potential applications of quantum dots.	5	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
